

Assignment 2 Flexible Aircraft Flight Dynamics

Guidelines:

We have learnt how to develop the linear flexible aircraft flight dynamic equations of motion. Firstly, we linearized the nonlinear rigid aircraft flight dynamics. Secondly, by using the mean body axes, the rigid-body degrees of freedom and structural degrees of freedom are decoupled inertially. Even so, they are still coupled aerodynamically. Thirdly, we learnt how to derive the aerodynamic Jacobian matrix of the horizontal and vertical tail. Finally, together with the aerodynamic coupling matrices, we have all the elements to assemble the linearized free-flying flexible aircraft flight dynamic equations.

This assignment aims at developing and analysing the linear flexible aircraft flight dynamics.

Model descriptions

We will use a glider model, whose wings are modelled as flexible cantilever beams. The horizontal and vertical tail of the glider is assumed to be rigid. The geometric and aerodynamic data can be found in the *Get_Glider_Properties.m* / *Get_Glider_Properties.py* file.

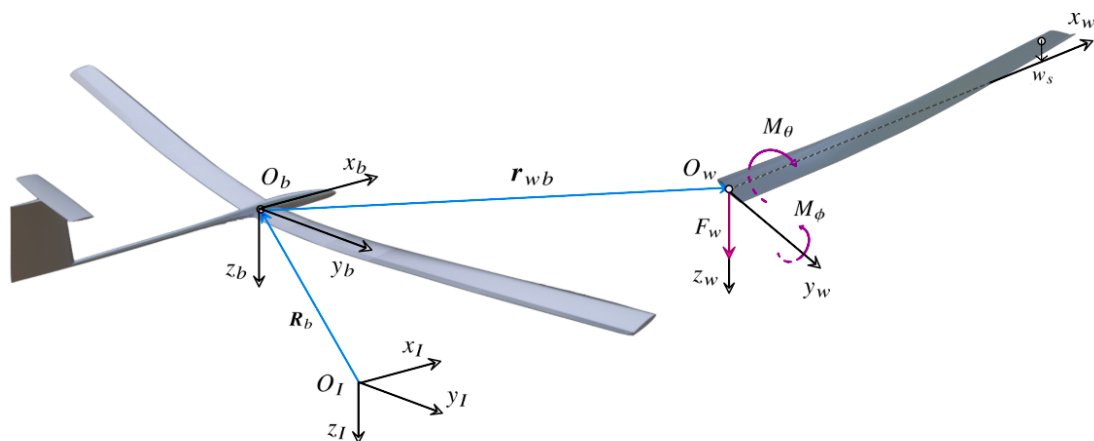


Figure 1 The glider model with flexible wings.

Clamped wing dynamic models

When assembling the state-space models for a clamped wing, we have different options. Regarding the aerodynamics, we can use either quasi-steady or unsteady strip theory. Regarding the structural model, we can either assume the wing as a rigid beam or include the structural flexibility.

The code for this part of the assignment has been fully given. No additional coding is required. You can directly run *main_Clamped_Wing_Dynamics.m/.py*, which calls the following functions:

1. *Get_FEM_Beam_Structure.m/.py*
2. *Get_Beam_Strip_Aerodynamics.m/.py*

3. Get_State_Space_Elastic_Wing.m/.py
4. Get_State_Space_Rigid_Wing.m/.py

Please follow these steps:

1. Analysis the state-space model for a rigid clamped beam. The only structural degree of freedom is the flap input beta. Plot the corresponding eigenvalues to the complex plane when using quasi-steady aerodynamics. Now plot the corresponding eigenvalues when using unsteady aerodynamics. Discuss your observations.
2. Analysis the state-space model for a flexible clamped beam. The finite element method is used for the structural model. Each node has four degrees of freedom: heave, out-of-plane bending, torsion, and flap input. Plot the corresponding eigenvalues to the complex plane when using quasi-steady aerodynamics. Now plot the corresponding eigenvalues when using unsteady aerodynamics. Discuss your observations.

Flight dynamic models

In this section, we will assemble all the elements together to obtain the fully coupled free-flying flight dynamic model for **a rigid aircraft with quasi-steady aerodynamics** and **a flexible aircraft with unsteady aerodynamics**.

The main code file needed is main_Aircraft_Flight_Dynamics.m/.py, which calls the following functions:

1. Get_Glider_Properties.m/.py (completed, no coding needed)
2. Get_Aero_Jacobian_Tail.m/.py (fill in the lines marked with to be decided (%tbd), using the knowledge learnt in class)
3. Get_EQM_Rigid_Dof.m/.py (completed, no coding needed)
4. Get_Coupling_Matrices.m/.py (fill in the lines marked with to be decided (%tbd), using the knowledge learnt in class)
5. Get_EQM_Coupled_Flight_Dynamics.m/.py (completed, no coding needed)

Once the files have been completed, follow the steps:

1. Plot the eigenvalues of **a rigid aircraft with quasi-steady aerodynamics** and **a flexible aircraft with unsteady aerodynamics** in the complex plane.
2. Regarding the rigid aircraft: Is the rigid aircraft stable? What are the rigid-body modes that you can identify?
3. Regarding the flexible aircraft: Is the flexible aircraft stable? What are the rigid-body modes that you can identify?
4. If you compare the eigenvalues of the rigid and flexible aircraft, what are the differences that you can observe? What are the impacts of flexibility? Do you see any new mode caused by the couplings?

Time domain simulations

By far, we have developed the state-space models for a **rigid aircraft with quasi-steady aerodynamics** and a **flexible aircraft with unsteady aerodynamics**. In this section, we will do time-domain simulations for comparing the dynamic responses of the two aircraft.

Remember the sampling frequency should be sufficiently high to capture the fast structural modes. The suggested settings are: *ode4 Runge-Kutta solver, 20 kHz sampling frequency, simulation time length 8 seconds*. Be careful about the conversions between radius and degree.

If you use Simulink, then you can directly run `Sim_Flight_Dynamics.slx`, which uses the initialization function `initialization.m`. If you use Python, do not worry, you can still use `initialization.py`, which has been fully coded for you. You just need to code a few lines to run the simulation and show the plots in Python.

We will consider six simulation cases:

1. Check the responses of the states when all the inputs equal zero. This is a nice step to get started. If all the settings are appropriate, you will be able to see all the states have zero responses throughout the simulation time length.
2. Elevator pulse input responses. The elevator deflects to 5 deg at $t = 1s$, and then deflects back to zero at $t = 3s$. All the rest inputs are zero.
3. Rudder pulse input responses. The rudder deflects to 5 deg at $t = 1s$, and then deflects back to zero at $t = 3s$. All the rest inputs are zero.
4. Flap pulse input responses. On the right wing, the hinge control moments of the two flaps on the tip increase to 10 Nm at $t = 1s$, and then go back to zero at $t = 3s$. Also, on the left wing, the hinge control moments of the two flaps on the tip decreases to -10 Nm at $t = 1s$, and then go back to zero at $t = 3s$. This case is similar to an aileron asymmetrical input. All the rests inputs are zero.
5. Responses to a “1-cos” gust. Just keep one period of the gust input, but run the simulation longer such that we can also see the aircraft responses after the gust is gone. The gust is uniform along the wing span. The maximum gust angle of attack input is 5 deg. The gust frequency is 0.2 Hz. All the rests inputs are zero.
6. Responses to a sharper “1-cos” gust. Just keep one period of the gust input, but run the simulation longer such that we can also see the aircraft responses after the gust is gone. Keep the maximum gust angle of attack input as 5 deg. Increase the gust frequency to 0.5 Hz. All the rests inputs are zero.

For each one of the simulation cases, please:

1. Plot the rigid-body states $[u \ v \ w \ p \ q \ r]$ for the rigid aircraft and the flexible aircraft, respectively. Describe your observations, especially on the differences between the rigid and flexible aircraft responses.
2. For the flexible aircraft, plot the nodal bending displacements and torsion angles. Remember, if the input is asymmetric, the responses on the left and right wings will also be different.

References

The following supporting materials can be downloaded from Brightspace.

1. Lecture slides.
2. Flow chart for the flight dynamic model in lecture slides.
3. Guideline for the Aerodynamic Jacobian of the Tail.pdf.
4. Supporting Matlab/Simulink code: AE4ASM522_Assignment2_Code_Matlab.zip.
5. Supporting Python code: AE4ASM522_Assignment2_Code_Python.zip.

Notes

- We encourage you to work in groups of 2-3.
- Please submit a group assignment report.
- Please also submit the code of the group for plagiarism check.
- Please write an individual contribution paragraph.